

Rising Waters

Flood Management System — AI/ML Project

Define Problem Statements

Phase 1: Brainstorming & Ideation

IBM Cloud

XGBoost 96.55%

Skill Wallet

AI/ML Track

1. Problem Statement

"How might we provide disaster management authorities and meteorologists with a reliable, real-time, machine-learning-powered flood prediction system that can accurately classify flood risk based on meteorological data — enabling timely evacuations and saving lives?"

2. Key Stakeholders

Meteorologists	Need accurate early warnings to issue advisories
Disaster Mgmt Teams	Need flood risk data to mobilise resources
Local Authorities	Need predictions to trigger evacuation protocols
Communities	Need timely warnings to evacuate safely
Government Bodies	Need data for infrastructure planning and policy

3. Root Cause Analysis

- Conventional forecasting methods lack ML-powered prediction accuracy
- Insufficient integration of multiple meteorological parameters
- No accessible web interface for non-technical disaster management users
- Delayed information flow between weather services and emergency responders

4. Desired Outcome

A publicly accessible web application that: (1) accepts 6 meteorological inputs, (2) preprocesses and scales the data, (3) runs an XGBoost classification model achieving 96.55% accuracy, and (4) presents a clear Flood Chance or No Flood Chance result with emergency action recommendations.